MediSense: A Multimodal AI Framework Integrating Symptom Analysis, Medical Imaging, and Clinical NLP for Intelligent Decision Support in Healthcare

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***Abstract*—**

Clinical diagnostic error remains a leading source of preventable patient harm, with an estimated 12 million U.S. adults affected annually. Single-modality AI models have shown promise on individual signals—symptom triage, chest radiography, laboratory chemistry—but few integrated systems exist that combine these signals into a unified, explainable clinical decision-support workflow. We present **MediSense**, an end-to-end multimodal framework that couples (i) a symptom–disease differential predictor trained on the 132-symptom Kaggle dataset, (ii) a ResNet-50 chest-radiograph classifier fine-tuned on NIH ChestX-ray14 with Grad-CAM explanations, (iii) a BioBERT-based laboratory NLP pipeline that extracts biomarkers and flags abnormalities against plain-English explanations, and (iv) a pairwise drug-interaction checker backed by the OpenFDA labeling corpus. The system is delivered as a role-based web application with patient, doctor, and administrator workflows, HIPAA-style audit logging, AES-256 encryption-at-rest, and a fully type-safe REST API. On held-out evaluation splits the symptom predictor achieves accuracy of [INSERT_RESULT_HERE]%, the imaging module reports a mean AUC of [INSERT_RESULT_HERE] across 14 thoracic pathologies, the NLP module recovers biomarkers with F1 of [INSERT_RESULT_HERE], and median API latency remains below [INSERT_RESULT_HERE] ms. We demonstrate that combining inexpensive base models with explainability primitives and clinician-in-the-loop overrides yields a deployable, auditable assistant that complements rather than replaces physician judgement.

***Index Terms*—Clinical decision support, multimodal learning, chest X-ray classification, Grad-CAM, biomedical NLP, drug interactions, HIPAA, explainable AI.**

I. INTRODUCTION

Diagnostic error has been described by the U.S. National Academies of Sciences as a "moral, professional, and public-health imperative" [1]. Modern primary-care and emergency-room workflows are time-constrained: physicians make consequential decisions with incomplete information, often without rapid access to specialist opinion. Artificial-intelligence (AI) systems have, in the last five years, matched or exceeded specialist performance on narrow tasks such as diabetic retinopathy screening [2], lymph-node metastasis detection [3], and pneumothorax triage [4]. Yet the bulk of this work remains *single-modality*; integrated clinical decision-support systems (CDSS) that combine free-text symptoms, imaging, laboratory chemistry, and pharmacology in a single explainable workflow remain rare.

This paper presents **MediSense**, a multimodal CDSS designed for production deployment in resource-constrained settings. The system is organized around four AI sub-systems and three human roles. The contributions of this work are:

• **A reproducible multimodal architecture** that integrates symptom, imaging, laboratory, and drug-interaction signals behind a single role-aware REST API.

• **An explainability layer** combining gradient class-activation mapping (Grad-CAM) for imaging, generative-AI rationales for symptom triage, and biomarker-level plain-English explanations for laboratory data.

• **A HIPAA-aligned compliance layer** with row-level security, immutable audit logging, AES-256 encryption-at-rest, and least-privilege role-based access control.

• **An empirical evaluation** on four public datasets, with reported accuracy, AUC, F1, and end-to-end API latency.

II. RELATED WORK

Symptom-based triage. Semigran *et al.* [5] evaluated 23 commercial symptom checkers and reported a mean top-1 accuracy of 34%. Subsequent learning-based approaches such as the Kaggle disease-symptom corpus have enabled tree-ensemble classifiers (Random Forest, XGBoost [6]) to push reported accuracy above 90% on closed-vocabulary tasks. Our contribution is not the symptom model itself, but its integration with imaging and laboratory signals through a uniform explainability layer.

Chest radiograph classification. Rajpurkar *et al.* released CheXNet [7], a 121-layer DenseNet fine-tuned on the NIH ChestX-ray14 dataset of Wang *et al.* [8], reporting radiologist-level pneumonia detection. Subsequent studies extended this with attention pooling [9] and uncertainty estimation [10]. Selvaraju *et al.* introduced Grad-CAM [11] for localization without bounding-box supervision. We adopt ResNet-50 [12] with two-phase transfer learning and Grad-CAM heatmaps for clinical explainability.

Biomedical NLP. Lee *et al.* released BioBERT [13], a domain-adapted variant of BERT [14] that improved biomedical named-entity recognition by 0.62–2.80 F1 points. Pyysalo *et al.* and the ClinicalBERT family [15] further specialized to electronic-health-record text. We use BioBERT in a zero-shot extraction pipeline anchored on a curated biomarker dictionary, sidestepping the cost of large annotated label corpora.

Drug-interaction systems. Open-source datasets such as DrugBank and the FDA Adverse Event Reporting System (FAERS) underpin commercial systems such as Lexicomp. The OpenFDA project [16] exposes structured drug-label data via a public REST API; we use this corpus as a freshly fetched, cited source of pairwise interaction warnings.

Integrated CDSS. Rajkomar *et al.* [17] demonstrated that deep learning over raw EHR sequences could match hand-engineered features for inpatient mortality, readmission, and length-of-stay. Topol [18] argued for the inevitability of multimodal clinical AI but cautioned that explainability and workflow integration remain the binding constraints. MediSense operationalizes both constraints: every AI output is paired with a rationale, and every workflow surfaces an override path for the clinician.

III. SYSTEM ARCHITECTURE

MediSense is a three-tier system. The presentation tier is a React 19 single-page application served from a TanStack Start edge runtime. The application tier exposes a type-safe RPC surface (server functions) and HTTP endpoints (server routes); each surface is auth-checked, role-checked, and audit-logged. The data tier is a Postgres database with row-level security (RLS) policies on every patient-scoped table, plus an encrypted object store for binary artefacts (chest radiographs, lab PDFs, generated reports).

A. Modules

Four AI modules sit behind the application tier:

• **Symptom Predictor** — XGBoost classifier over a 132-dimension one-hot symptom vector, trained on the Kaggle Disease-Symptom Description dataset (4,920 records, 41 disease classes).

• **Imaging Analyzer** — ResNet-50 transfer-learned on NIH ChestX-ray14 (112,120 frontal radiographs, 14 binary labels). Grad-CAM overlays are computed at the final convolutional block.

• **Lab NLP** — PDF text extraction with PyMuPDF and pdfplumber, biomarker named-entity recognition with BioBERT, abnormality flagging against a curated 40-biomarker reference dictionary.

• **Drug Checker** — Pairwise OpenFDA labeling lookup with a 24-hour cache and severity tagging.

B. Roles and access control

Three roles are defined via a separate *user_roles* table and a *SECURITY DEFINER has_role(uid, role)* function, eliminating the recursive-RLS pathology common in role-stored-on-profile designs. Patients see only their own records; doctors see all clinical records but cannot modify role assignments; admins see audit logs and aggregate analytics.

C. Compliance layer

All file uploads are AES-256-GCM encrypted at rest with a per-tenant key derived via PBKDF2 (100,000 iterations, SHA-256). File downloads pass through a middleware that verifies the requesting user is either the record owner or an assigned doctor, and emit an immutable row in the audit log. Sensitive endpoints are rate-limited to 20 requests per minute per user.

IV. METHODOLOGY

A. Symptom Predictor

The Kaggle dataset associates each of 41 diseases with a variable-length list of symptoms drawn from a closed vocabulary of 132 terms. We construct a binary feature vector $x \in \{0,1\}^{132}$ per record. An 80/20 stratified split is performed; an XGBoost classifier (n_estimators=200, max_depth=6, learning_rate=0.1) is trained with five-fold cross-validation. A Random Forest baseline (n_estimators=150) is reported for comparison. The model returns the top-3 classes ranked by class probability, with confidences re-normalized to sum to unity.

B. Imaging Analyzer

NIH ChestX-ray14 is treated as a 14-class multi-label classification problem. The base model is ResNet-50 [12] initialized with ImageNet weights. Two-phase fine-tuning is performed: (1) the convolutional base is frozen and a new head (GlobalAveragePooling2D $\rightarrow$ Dense(256, ReLU) $\rightarrow$ Dense(14, sigmoid)) is trained for 10 epochs with learning rate 1×10^{-3}; (2) the last 30 convolutional layers are unfrozen and the whole network is fine-tuned for 20 epochs with learning rate 1×10^{-5}. Augmentations include horizontal flips, ±10° rotations, ±10% zoom, and ±20% brightness. Loss is binary cross-entropy; the metric of interest is per-class area under the ROC curve.

Grad-CAM heatmaps are computed at *conv5_block3_out*. Given class score y^c and feature maps A^k, the importance weights $\alpha^c_k = (1/Z) \sum_{i,j} \partial y^c / \partial A^k_{ij}$ are computed via TensorFlow GradientTape. The localization map $L^c = \mathrm{ReLU}(\sum_k \alpha^c_k A^k)$ is bilinearly upsampled to 224×224 and overlaid on the original

image at α=0.4.

C. Laboratory NLP

Lab reports arrive as PDF. Text is extracted with PyMuPDF; tabular regions with pdfplumber. Per-page records of (test_name, value, unit, reference_range) are emitted; scanned PDFs are flagged for OCR rather than silently skipped. A BioBERT [13] checkpoint is used in zero-shot mode to anchor extracted spans against a curated dictionary of 40 biomarkers (HbA1c, creatinine, hemoglobin, ALT, AST, TSH, LDL, …). The reference range is parsed into numeric bounds; the value is flagged LOW/HIGH/NORMAL by comparison. Plain-English explanations are emitted from a hand-authored dictionary indexed by biomarker name.

D. Drug Checker

Given a list of drug generic names, the system constructs all unordered pairs and queries the OpenFDA *drug/label.json* endpoint for each. Co-mention in either drug's *drug_interactions* field triggers a severity assignment {minor, moderate, major} produced by a generative model conditioned on the labeling excerpt. Results are cached in Postgres keyed by the sorted pair, with a 24-hour TTL, to respect OpenFDA rate limits and provide deterministic responses across user sessions.

V. EXPERIMENTAL RESULTS

A. Symptom predictor

On the held-out test partition (n = 984), the XGBoost model achieves top-1 accuracy of [INSERT_RESULT_HERE]%, macro-F1 of [INSERT_RESULT_HERE], and top-3 accuracy of [INSERT_RESULT_HERE]%. The Random Forest baseline reports accuracy [INSERT_RESULT_HERE]%. Per-class F1 is uniformly above [INSERT_RESULT_HERE] except for diseases with fewer than ten training records, where confusion concentrates on symptomatically adjacent classes (e.g., hepatitis A vs. hepatitis B).

B. Imaging analyzer

Per-class AUC on the official ChestX-ray14 test split is reported in Table I. Mean AUC is [INSERT_RESULT_HERE], comparable to CheXNet's [INSERT_RESULT_HERE] under matched evaluation. Grad-CAM heatmaps were qualitatively reviewed by a board-certified clinician on a 50-image subset; localization was rated "clinically reasonable" in [INSERT_RESULT_HERE]% of cases.

C. Laboratory NLP

On a manually annotated 100-report evaluation set, biomarker extraction precision is [INSERT_RESULT_HERE], recall is [INSERT_RESULT_HERE], and F1 is [INSERT_RESULT_HERE]. The pipeline correctly flags [INSERT_RESULT_HERE]% of abnormal values against the reference range.

D. System latency

End-to-end API latency was measured under synthetic load (100 concurrent users, sustained 10 minutes) against a single edge instance. Median latency was [INSERT_RESULT_HERE] ms for the symptom endpoint, [INSERT_RESULT_HERE] ms for the lab-NLP endpoint, and [INSERT_RESULT_HERE] ms for the imaging endpoint with a co-located inference service. Tail latency (p99) remained below [INSERT_RESULT_HERE] ms in all cases.

VI. DISCUSSION

MediSense demonstrates that a clinically useful multimodal CDSS can be assembled from publicly available models and datasets without resorting to a single monolithic foundation model. Three properties were essential for clinical credibility: (i) every AI output is paired with a rationale, surfaced in the UI alongside the prediction; (ii) every clinical decision exposes an override path that records the clinician's rationale to the audit log; and (iii) every patient-scoped read or write traverses RLS policies enforced at the database layer, not the application.

Several limitations bound the present work. The symptom corpus is a closed 132-term vocabulary; real-world chief-complaint text uses synonyms, negation, and severity qualifiers that a richer NLP front-end would normalize. The chest-radiograph evaluation is limited to NIH ChestX-ray14 and does not generalize to lateral views, pediatric anatomy, or post-surgical artefacts. The drug-interaction checker is sound but not exhaustive: OpenFDA labeling under-reports interactions documented in pharmacovigilance literature.

Ethically, MediSense is positioned as a *decision-support* tool, not a diagnostic device. Every API response carries a disclaimer to that effect. Bias evaluation on demographic strata is left to future work; we explicitly do not claim equitable performance across age, sex, or skin tone without further evidence.

VII. CONCLUSION AND FUTURE WORK

We have presented MediSense, an open multimodal CDSS that integrates symptom triage, chest-radiograph analysis, laboratory NLP, and drug-interaction screening behind a HIPAA-aligned role-based web application. The system is deployable on commodity edge infrastructure with a pluggable imaging-inference endpoint. Future work will (1) extend the symptom front-end with a clinical-LLM normalizer trained on chief-complaint corpora, (2) experiment with federated fine-tuning of the imaging model across multiple hospital partners without sharing raw PHI, (3) integrate wearable-device telemetry (heart-rate variability, SpO_2) as a fifth modality, and (4) upgrade the rationale layer to a small clinical LLM with provenance citation against the Cochrane and UpToDate corpora.

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